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Thermal cycler systems and methods of use

Abstract

A thermal cycler system comprises a sample block configured to receive a sample holder configured to receive a plurality of samples; an adaptor configured to surround a periphery of the sample block; and a drip pan configured to surround a periphery of the adaptor. The drip pan comprises one or more ejector mechanisms and one or more openings, the one or more ejector mechanisms configured to respectively extend through the one or more openings into contact with the sample holder in a state of the sample holder received by the sample block and the adaptor surrounding the periphery of the sample block.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6340589	12/2001	Turner et al.	N/A	N/A
6719949	12/2003	Barzilai et al.	N/A	N/A
10471432	12/2018	Tan	N/A	B01L 7/52
11548007	12/2022	Tan	N/A	B01L 7/52
11944975	12/2023	Tan	N/A	B01L 9/523
2002/0030044	12/2001	Brown	N/A	N/A
2003/0044969	12/2002	Shin	435/303.1	B01L 3/5085
2005/0084957	12/2004	Atwood et al.	N/A	N/A
2007/0184548	12/2006	Tan et al.	N/A	N/A
2007/0212775	12/2006	Cohen et al.	N/A	N/A
2010/0028988	12/2009	Chu	264/291	B29C 51/04
2010/0124766	12/2009	Ng	435/303.1	B01L 7/52
2012/0145587	12/2011	Yeo	206/562	B01L 7/52
2015/0231636	12/2014	Kiong et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103415346	12/2012	CN	N/A
103857464	12/2013	CN	N/A
204874536	12/2014	CN	N/A
10115848	12/2001	DE	B01L 7/00
102005044021	12/2006	DE	N/A
102011109332	12/2012	DE	N/A
2001/56697	12/2000	WO	N/A
2012078930	12/2011	WO	N/A
2015/200788	12/2014	WO	N/A

OTHER PUBLICATIONS

European Communication issued in Application No. 16 823 510.9, dated Jul. 23, 2019. cited by applicant
Singapore Search Report and Written Opinion issued in Application No. 11201805240P, dated Aug. 14, 2019. cited by applicant
PCT/U82016/068151, Partial International Search Report dated Mar. 14, 2017, 3 pages. cited by applicant
PCT/US2016/068151, International Search Report and Written Opinion dated May 12, 2017, 9 pages. cited by applicant
Singapore Examination Report issued in corresponding Application No. 11201805240P, dated Dec. 22, 2020. cited by applicant
Chinese Office Action issued in corresponding Application No. 201680078409.2, dated Apr. 3, 2020. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 18/148,139, filed Dec. 29, 2022, which is a continuation application of U.S. application Ser. No. 16/590,459, filed Oct. 2, 2019 (now U.S. Pat. No. 11,548,007), which is a divisional application of U.S. application Ser. No. 15/387,614, filed Dec. 21, 2016 (now U.S. Pat. No. 10,471,432), which claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application No. 62/372,876 filed on Aug. 10, 2016 (now expired) and to U.S. Provisional Patent Application No. 62/270,716 filed on Dec. 22, 2015 (now expired), each of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD
(1) The present invention relates generally to thermal cycler systems and methods of using same.
INTRODUCTION
(2) Testing of biological or chemical samples often requires a device for repeatedly subjecting multiple samples though a series of temperature cycles. To prepare, observe, test, and/or analyze an array of biological samples, one example of an instrument that may be utilized is a thermal cycler or thermocycling device, such as an end-point polymerase chain reaction (PCR) instrument or a quantitative, or real-time, PCR instrument. Such devices are used to generate specific temperature cycles, i.e. to set predetermined temperatures in the reaction vessels to be maintained for predetermined intervals of time.
(3) Generally, thermal cycler systems include a sample block that has a plurality of reaction regions or sample block wells and that is configured to receive a plurality of samples contained in sample wells of a sample holder. The samples may be sealed within the wells of the sample holder via a lid, cap, sealing film or any other sealing mechanism between the wells and a heated cover. A variety of sample holders are used in thermal cycler systems including, for example, a multi-well microtiter plate, a micro card, or a through-hole array. Due to the variety of available sample holders, thermal cycler systems are often designed to be compatible with more than one type of sample holder. For example, sample blocks may be configured to receive a sample holder having either a full skirt or a semi-skirt. A full-skirted sample holder has skirting that generally extends on at least two opposite sides of the sample holder to the bottom portions of the sample wells, while

the skirting of the semi-skirted sample holder leaves lower portions of the sample wells exposed. Designing a thermal cycler system compatible with sample holders having different designs often leads to inefficiencies depending on the actual sample holder used. To perform the PCR process successfully, efficiently, and accurately, these inefficiencies should be minimized to the greatest extent possible.
(4) There is an increasing need to provide improved thermal cycler systems that address one or more of the above drawbacks.
SUMMARY
(5) In accordance with one embodiment, a thermal cycler system for use with a sample holder configured to receive a plurality of samples includes a sample block having an upstanding peripheral side wall and being configured to receive the sample holder and an adaptor having an upstanding peripheral side wall configured to be positioned about the peripheral side wall of the sample block. When the peripheral side wall of the adaptor is positioned about the peripheral side wall of the sample block and the sample holder is received in the sample block, the peripheral side wall of the adaptor extends in an upward direction toward the sample holder.
(6) In accordance with another embodiment, an adaptor configured to be positioned about a sample block, the sample block including an upstanding peripheral side wall and being configured to receive a sample holder, includes an upstanding peripheral side wall. When the peripheral side wall of the adaptor is positioned about the peripheral side wall of the sample block and the sample holder is received in the sample block, the peripheral side wall of the adaptor extends in an upward direction toward the sample holder.
(7) Various additional features and advantages of the invention will become more apparent to those of ordinary skill in the art upon review of the following detailed description of the illustrative embodiments taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with a general description of the invention given above, and the detailed description given below, serve to explain the invention.
(2) FIG. **1** is a perspective view of a thermal cycler system according to one embodiment showing an adaptor and sample holder positioned about the sample block.
(3) FIG. **2** is an exploded view of the thermal cycler system of FIG. **1** showing the sample holder removed from the sample block.
(4) FIG. **3** is an exploded view of the thermal cycler system of FIG. **1** showing the adaptor and insulation components removed from the sample block without the sample holder and showing a portion of the housing in cross-section.
(5) FIG. **4**A is a cross-sectional view of a portion of the thermal cycler system of FIG. **1** and the sample holder.
(6) FIG. **4**B is a cross-sectional view of the portion of the thermal cycler system of FIG. **4**A showing the sample holder positioned on the adaptor and the sample block.
(7) FIG. **5** is a cross-sectional view of a portion of the thermal cycler system of FIG. **1** and a sample holder having a different design than the sample holder of FIG. **4**A.
(8) FIG. **6**A is a perspective view of a sample block above a drip pan with wire form springs.
(9) FIG. **6**B is a lengthwise side view of the sample block and drip pan of FIG. **6**A.
(10) FIG. **6**C is a widthwise side view of the sample block and drip pan of FIG. **6**A.
DETAILED DESCRIPTION
(11) Referring to FIGS. **1-3**, a thermal cycler system **10** is shown constructed in accordance with an

illustrative embodiment of the present invention. The thermal cycler system **10** includes an outer housing **12**, a sample block **14**, and a drip pan **16**. The sample block **14** includes a plurality of cavities **18** and is configured to be loaded with a correspondingly shaped sample holder **20** containing a plurality of biological or biochemical samples in a plurality of sample wells **22**. The drip pan **16** is designed to seal components of the thermal cycler system **10**, such as a thermal block assembly (not shown), from environmental conditions above the drip pan **16**. A thermal block assembly may include, for example, a heating and cooling element and a heat exchanger or heat sink for heating and cooling the biological or biochemical samples during the PCR process. The thermal cycler system **10** is described in greater detail below.

(12) Still referring to FIGS. **1-3**, the thermal cycler system **10** includes an access area **24** for the sample holder **20** to be inserted and removed. In various embodiments, the access area **24** is configured to include enough open space for a robotic arm of a lab automation system (not shown) to position the sample holder **20** on the sample block **14**. Additionally, the thermal cycler system **10** is configured to be compatible with a full-skirted sample holder (not shown). The space required for the manipulation of the sample holder **20** by a robotic arm poses a problem when the sample holder is semi-skirted rather than full-skirted. In that regard, when a semi-skirted sample holder **20** is received by the sample block **14**, a peripheral wall **26** of the sample block **14** is exposed. Thus, the sample block **14** is vulnerable to an external air draft, which may severely affect the consistent thermal performance of the thermal cycler system **10**. Accordingly, in one embodiment, the thermal cycler system **10** includes an adaptor **28**, which is described in greater detail below.

(13) The exemplary thermal cycler system **10**, unless otherwise indicated, is described herein using a reference frame in which the sample holder **20** may be loaded in the front of the thermal cycler system **10** and may be positioned above the sample block **14**. Consequently, as used herein, terms such as lateral, forward, backward, downward, upward, beneath, and above used to describe the exemplary thermal cycler system **10** are relative to the chosen reference frame. The embodiments of the present invention, however, are not limited to the chosen reference frame and descriptive terms. Those of ordinary skill in the art will recognize that the descriptive terms used herein may not directly apply when there is a change in reference frame. Nevertheless, the relative terms used to describe embodiments of the thermal cycler system **10** are to merely provide a clear description of the embodiments in the drawings. As such, the relative terms lateral, forward, backward, downward, upward, beneath, and above are in no way limiting the present invention to a particular location or orientation.

(14) With reference now to FIGS. **3** and **4**A, the exemplary sample block **14** is shown in more detail. The sample block **14** includes a base **30** and the upstanding peripheral side wall **26**, which encloses the plurality of cavities **18**. As described above, the plurality of cavities **18** are configured to receive the plurality of correspondingly shaped sample wells **22** of the sample holder **20**. In the illustrative embodiment, the sample block **14** includes 96 cavities **18**. In such an embodiment, the sample holder **20** may be a 96-well microtiter plate. It should be recognized that the sample block **14** and the sample holder **20** may have alternate configurations. For example, the sample holder **20** may be, but is not limited to, any size multi-well plate, card or array including, but not limited to, a 24-well microtiter plate, 50-well microtiter plate, a 384-well microtiter plate, a microcard, a through-hole array, or a substantially planar holder, such as a glass or plastic slide.

(15) Still referring to FIGS. **3** and **4**A, the exemplary drip pan **16** is shown in more detail. The drip pan **16** forms a seal between the sample block **14** and the drip pan **16** to isolate thermoelectric components (not shown) from environmental conditions above the sample block **14** and the drip pan **16**. In particular, the drip pan **16** prevents any sample that may splash out of the sample wells **22** from reaching the sensitive electronic components of the thermal block assembly (not shown). The drip pan **16** includes a side wall **32** and a bottom surface **34**. In one embodiment, the drip pan **16** is configured to receive the adaptor **28**. Further, the drip pan **16** may be configured to secure a lateral position of the adaptor **28** relative to the drip pan **16**. In that regard, when the adaptor **28** is

received by the drip pan **16**, the side wall **32** of the drip pan **16** prevents lateral movement of the adaptor **28**.

(16) With reference again to FIGS. **3** and **4**A, the adaptor **28** is shown in more detail. The adaptor **28** includes a deck portion **36** including a plurality of apertures **38**. The plurality of apertures **38** is configured to allow the array of sample wells **22** of the sample holder **20** to extend therethrough when the adaptor **28** is positioned about the sample block **14** and sample holder **20** is received by the sample block **14** (shown in FIG. **4**B). A perimeter **40** of the deck portion **36** is formed with an upstanding peripheral side wall **42** extending downwardly beneath the deck portion **36**. The upstanding peripheral side wall **42** is configured to be positioned about the peripheral side wall **26** of the sample block **14**. When the peripheral side wall **42** of the adaptor **28** is positioned about the peripheral side wall **26** of the sample block **14** and the sample holder **20** is received in the sample block **14**, the peripheral side wall **42** of the adaptor **28** extends in an upward direction toward the sample holder **20** (described below). In other words, the peripheral side wall **42** of the adaptor **28** may extend in a direction from the base **30** of the sample block **14** towards a deck portion **44** of the sample holder **20**. In this manner, the peripheral side wall **42** of the adaptor **28** is configured to protect the peripheral side wall **26** of the sample block **14** from undesirable contact with air flow during the PCR process. It should be recognized that the peripheral side wall **42** of the adaptor **28** may be a continuous or a discontinuous side wall. In other words, in various embodiments, the peripheral side wall **42** may comprise one or more wall segments.

(17) Still referring to FIGS. **3** and **4**A, the perimeter of the upstanding peripheral side wall **42** of the adaptor **28** includes a lip **46** extending therefrom. The lip **46** is configured to be received by the drip pan **16**. When the lip **46** is received by the drip pan **16**, the lateral position of the adaptor **28** may be secured by the drip pan **16**. In one embodiment, an insulation component **47** may be positioned between the drip pan **16** and the lip **46** of the adaptor **28**. The insulation component **47** may be adhered to the drip pan **16**, for example. Additionally, in one embodiment, insulation components **49** may be coupled to the deck portion **36** of the adaptor **28**. The insulation components **49** may aid in preventing draft air from the front and back of the thermal cycler system **10**. In addition, the insulation components **49** may also act as a secondary uniform force on the bottom of the sample holder **20** to aid in the ejection of the sample holder **20** after the PCR process is complete. The insulation components **47**, **49** may be made of BISCO® HT-800 Medium Cellular Silicone available from Rogers Corporation in Rogers, Conn. for example. While the adaptor **28** is shown as including the deck portion **36** and the lip **46**, it should be recognized that other configurations of the adaptor **28** are possible. For example, an adaptor according to one embodiment may not include a deck portion or a lip. Further, in one embodiment, the adaptor **28** may be configured to accommodate a full-skirted sample holder (not shown). In an embodiment where the adaptor **28** is configured to accommodate only semi-skirted sample holders, the adaptor **28** may be positioned about the sample block **14** before the sample holder **20** is loaded. For subsequent runs, no user intervention or replacement is necessary until the user wants to use a full-skirted sample holder.

(18) Referring again to FIGS. **2** and **3**, in one embodiment, the drip pan **16** includes a plurality of ejector mechanisms **48**. While the illustrated embodiment shown in FIG. **3** depicts four ejector mechanisms **48**, other embodiments may employ a single ejector mechanism **48** or a suitable number of a plurality of ejector mechanisms **48**. The ejector mechanisms **48** may allow for easier removal of the sample holder **20** after the PCR process is complete. Each ejector mechanism **48** may comprise one or more springs that are compressed when a sample holder **20** is placed onto the sample block. As illustrated within the embodiments shown in FIGS. **2** and **3**, the springs are contained within a housing component of the ejector mechanisms **48**, but other embodiments may employ different housings or no housing at all. Additionally, the number and size of ejector mechanisms **48** (and the number of size of springs within ejector mechanisms **48**) will vary depending on the size and format of drip pan **16**, sample block **14**, sample holder **20** and any

adaptor **28** that is employed. To account for the ejector mechanisms **48**, the adaptor **28** includes a plurality of openings **50** configured to allow the ejector mechanisms **48** to extend therethrough in order to make contact with sample holder **20** for the purposes of ejection. The openings **50** may extend beyond the perimeter of the peripheral side wall **42**. Therefore, the peripheral side wall **42** may include extensions **52**. When the ejector mechanisms **48** extend through the openings **50**, the extensions **52** of the peripheral side wall **42** at least partially surround the ejector mechanisms **48**. Further, the perimeter **40** of the deck portion **36** may extend inward from the perimeter of the peripheral side wall **42**. In the illustrated embodiment, the openings **50** extend across a section of the deck portion **36** that may otherwise include apertures **38**. Accordingly, the openings **50** may be configured to allow one or more of the sample wells **22** of the sample holder **20** to extend therethrough when the sample holder **20** is positioned adjacent the adaptor **28**. Further, a deck portion segment **54** of the deck portion **36** may extend to the perimeter of the peripheral side wall **42** between the openings **50**. In this manner, the peripheral side wall **42** acts to protect the sample block **14** from undesirable contact with air flow while allowing the ejector mechanisms **48** to extend through the adaptor **28**.

(19) With reference to FIGS. **6**A-**6**C, an embodiment is illustrated where drip pan **16** includes wire form springs **68** as an embodiment of ejector mechanisms **48**. In such embodiments, the springs are not contained within a housing, as depicted within the illustrated embodiment shown in FIGS. **2-3**. Within the illustrated embodiment of FIGS. **6**A-**6**C, drip pan **16** includes a wire form spring **68** on each of its four sides around where the sample block is placed. Other embodiments may include more than one wire form spring **68** per side, or only include one or more wire form springs **68** on a subset of the sides of drip pan **16** (e.g., on one side, two sides, or three sides). Additionally, one or more wire form springs **68** may be employed in combination with other ejector mechanisms **48**, such as but not limited to those described in association with the embodiments illustrated in FIGS. **2** and **3**.

(20) With further reference to FIGS. **6**A-**6**C, embodiments employing wire form springs **68** may be configured to operate without an adaptor **28**, with sample wells **22** being inserted into cavities **18** when sample holder **20** is placed on sample block **14**. The wire form springs **68** are located on drip pan **16** such that sample holder **20** is placed on top of wire form springs **68**, which compresses the wire form springs **68**. In this fashion, wire form springs **68** can assist in ejection of sample holder **20**. The use of wire form springs **68** can be beneficial when spatial constraints may not allow the use of other ejector mechanisms **48** on one or more sides of drip pan **16**, or when the spatial constraints do not allow the use of an adaptor **28** (e.g., when spatial constraints do not allow the use of an adaptor **28** with a plurality of openings **50** through which ejector mechanisms **48** extend, as illustrated within the embodiment shown in FIG. **3**). Certain embodiments, including but not limited to those using an adaptor **28**, may combine wire form springs **68** with other ejector mechanisms **48** to enhance the overall ejection of sample holder **20**. Wire form springs **68** can comprise any suitable material. Non-limiting examples of suitable wire form springs **68** include music wire of 0.90 mm of SWP-B, JIS G3522 with zinc plating or chromium finishing, and also stainless steel wire springs of 0.90 mm of stainless steel 17-7 PH with precipitation hardening. Other suitable materials for wire form springs **68** include high carbon steel, carbon alloys, hard drawn steel, steel alloys, non-ferrous alloys, high temperature alloys, and other metals and alloys known in the art.

(21) With further reference to FIGS. **2** and **3**, embodiments employing a sample holder **20** in a full skirt configuration may utilize the heated cover and adaptor **28** to enhance removal of sample holder **20**. In such embodiments, when the heated cover is lowered to provide a downward force to the sample holder **20** as discussed below in reference to FIGS. **2** and **4**A, the skirt of sample holder **20** will sit on top of and be depressed into a portion of adaptor **28**. The materials for the skirt of sample holder **20** and the adaptor **28** are chosen in such embodiments to allow the skirt to be depressed into the adaptor without damaging either component. For example, a skirt wall **62** of

plastic and the corresponding portion of adaptor **28** of silicon rubber allowing for repeated use with the plastic skirt wall **62** being depressed into the silicon rubber portion of adaptor **28**. Any appropriate portion of the skirt of sample holder **20** can be depressed into adaptor **28**, such as a side or sides of sample holder **20** that do not interact with other features (for example, ejector mechanisms **48**) to enhance removal of sample holder **20**. Removal of the heated cover removes the downward force onto sample holder **20**, thereby creating a spring cantilever force to eject sample holder **20**. In one embodiment, sample holder **20** in a full skirt configuration employs the depression of the skirt into adaptor **28** for the ejection and removal of sample holder **20**. In another embodiment, the full skirt configuration of sample holder **20** being depressed into adaptor **28** is combined with the use of ejector mechanisms (for example, the plurality of ejector mechanisms as described in the illustrated embodiment within FIG. **3**). In embodiments where drip pan **16** includes one or more ejector mechanisms **48** and a full skirt configuration of sample holder **20** is utilized, the skirt may be depressed into adaptor **28** along sides or portions for which there are no ejector mechanisms **48**, thereby providing ejection force (from, for example, both or either ejector mechanisms **48** or from the spring cantilever force created when the heated cover is lowered onto the sample holder) that will act on multiple sides of sample holder **20**. In reference to the illustrated embodiment of FIG. **3**, use of a sample holder **20** with a full skirt would allow the use of ejector mechanisms **48** along the short sides of sample holder **20** to be combined with depression of the long sides of the skirt into adaptor **28** in order to provide ejection force on all four sides of sample holder **20**. Embodiments utilizing the creation of a spring cantilever force to aid in removal of sample holder **20** after lifting of the heated cover provide advantages in ensuring complete removal. The temperatures involved during thermal cycling can complicate complete removal as the heat can cause thermal warpage of sample holder **20**, such as when higher temperatures during thermocycling are employed or when sample holder **20** comprises non-hard shell materials that are more susceptible to thermal warpage.

(22) With further reference to FIG. **3**, in one embodiment, the drip pan **16** and the adaptor **28** include corresponding mating features. The corresponding mating features act as a self-locating feature to ensure the proper placement of the adaptor **28**. In the illustrated embodiment, the side wall **32** of the drip pan **16** includes projections **56**, and the lip **46** of the adaptor **28** includes recesses **58**. The projections **56** are configured to engage the recesses **58** when the adaptor **28** is received by the drip pan **16**. In that manner, the adaptor **28** is unlikely to be displaced if it is accidently hit by a robotic arm (not shown) during operation.

(23) With reference now to FIGS. **2** and **4**A, the exemplary sample holder **20** is shown in more detail. The sample holder **20** includes a deck portion **44** that supports the plurality of sample wells **22** in a regular array or matrix. The deck portion **44** serves to connect the adjacent sample wells **22** near to or at the top of each sample well **22** and to hold them in the desired matrix. The sample wells **22** are designed with generally thin walls to allow heat transfer to take place between the sample block **14** and the contents of the well. A perimeter **60** of the deck portion **44** is commonly formed with a skirt wall **62** extending downwardly beneath the deck portion **44**. The skirt wall **62** may be integrally formed with the deck portion **44** during molding of the sample holder **20** and generally forms a continuous wall of constant height around the sample holder **20**. In the illustrated embodiment, the sample holder **20** is semi-skirted meaning the skirt wall **62** does not extend to the bottom of the sample wells **22**. The skirt wall **62** lends stability to the sample holder **20** when it is placed on a surface and some rigidity when the sample holder **20** is being handled. The sample holder **20** is configured to be positioned over the sample block **14** and the adaptor **28**. A heated cover (not shown) may provide a downward force to the sample holder **20**. The downward force provides vertical compression between the sample holder **20**, the sample block **14**, and the other components of thermal block assembly (not shown), which improves thermal contact between the sample block **14** and the sample holder **20** to heat and cool the samples in the sample wells **22**. The heated cover may also prevent or minimize condensation and evaporation above the samples

contained in the sample wells **22**, which can help to maintain optical access to samples.

(24) Referring again to FIGS. **3** and **4**A, the sample wells **22** of the sample holder **20** are configured to receive a plurality of samples. The sample wells **22** may be sealed within the sample holder **20** via a lid, cap, sealing film or other sealing mechanism between the sample wells **22** and the heated cover (not shown). The sample wells **22** in various embodiments of a sample holder **20** may include depressions, indentations, ridges, and combinations thereof, patterned in regular or irregular arrays formed on the surface of the sample holder **20**. Sample or reaction volumes can also be located within wells or indentations formed in a substrate, spots of solution distributed on the surface a substrate, or other types of reaction chambers or formats, such as samples or solutions located within test sites or volumes of a microfluidic system, or within or on small beads or spheres. Samples held within the sample wells **22** may include one or more of at least one target nucleic acid sequence, at least one primer, at least one buffer, at least one nucleotide, at least one enzyme, at least one detergent, at least one blocking agent, or at least one dye, marker, and/or probe suitable for detecting a target or reference nucleic acid sequence.

(25) With reference to FIGS. **4**A and **4**B, the configuration of the sample block **14**, the adaptor **28**, and the sample holder **20** is shown in more detail. A user may position the adaptor **28** so that the peripheral side wall **42** of the adaptor **28** is positioned about the peripheral side wall **26** of the sample block **14**. Some of the cavities **18** of the sample block **14** are aligned with the apertures **38** of the adaptor **28**, while other cavities **18** are aligned with the openings **50** (not shown in the cross-section of FIG. **4**B). Next, the user may position the sample holder **20** on the sample block **14**. The sample wells **22** of the sample holder **20** extend through the apertures **38** or openings **50** of the adaptor **28** and into the cavities **18** of the sample block **14**. The deck portion **36** of the adaptor **28** may be configured to maintain proper engagement between the sample wells **22** of the sample holder **20** and the cavities **18** of the sample block **14**. For example, the thickness of the deck portion **36** of the adaptor **28** may be designed so as to allow the sample wells **22** to properly extend into the cavities **18**. If the thickness of the deck portion **36** is too large and the sample wells **22** are not properly engaged with the cavities **18**, the heat transfer between the sample wells **22** and the cavities **18** may be significantly impacted leading to process inefficiencies. When the sample holder **20** is received by the sample block **14**, the peripheral side wall **42** of the adaptor **28** extends in an upward direction toward the sample holder **20**. In other words, the peripheral side wall **42** of the adaptor **28** extends in a direction from the base **30** of the sample block **14** towards the deck portion **44** of the sample holder **20**. Further, the peripheral side wall **42** extends laterally in a space between the skirt wall **62** of the sample holder **20** and the peripheral side wall **26** of the sample block **14**. In this manner, the peripheral side wall **42** of the adaptor **28** protects the peripheral side wall **26** of the sample block **14** from undesirable air flow that would interfere with the heat transfer during the PCR process.

(26) Advantageously, the configuration of the adaptor **28** allows for the thermal cycler system **10** to be compatible with sample holders that vary in design. The design of the peripheral side wall of commercially available sample holders may vary, for example. With reference to FIG. **5**, a sample holder **64** is shown positioned on the adaptor **28**. As can be seen, the sample holder **64** has a design that differs from the design of the sample holder **20** shown in FIGS. **4**A and **4**B. Thus, the adaptor **28** is configured to receive a variety of sample holders.

(27) While the present invention has been illustrated by the description of specific embodiments thereof, and while the embodiments have been described in considerable detail, it is not intended to restrict or in any way limit the scope of the appended claims to such detail. The various features discussed herein may be used alone or in any combination. Additional advantages and modifications will readily appear to those skilled in the art. The invention in its broader aspects is therefore not limited to the specific details, representative apparatus and methods and illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the scope or spirit of the general inventive concept.

Claims

1. A thermal cycler system comprising: a sample block configured to receive a sample holder; one or more ejector mechanisms configured to contact the sample holder; and an adaptor configured to receive the sample block and the sample holder, wherein the adaptor comprises one or more of openings configured to allow the one or more ejector mechanisms to extend through the one or more openings, and wherein in a state of the sample block and the sample holder received by the adaptor, the one or more ejector mechanisms extend through the one or more openings and make contact with the sample holder.
2. The system of claim 1, the one or more ejector mechanisms are positioned adjacent one or more sides of the sample block, and wherein the one or more ejector mechanisms are configured to cause force to be exerted on the sample holder in a direction away from the sample block in a state of the sample holder received by the sample block.
3. The system of claim 1, wherein the adaptor is configured to removably receive the sample block, and wherein the adaptor further comprises an adaptor surface defining a plurality of apertures.
4. The system of claim 3, wherein the one or more openings are configured to receive the one or more ejector mechanisms in a position to be placed in contact with the sample holder supported on the surface.
5. The system of claim 4, wherein, in a state of the sample block received by the adaptor, the plurality of apertures are positioned over a plurality of cavities, respectively.
6. The system of claim 4, wherein: the adaptor further comprises a peripheral sidewall extending from the surface, and the adaptor surface and peripheral sidewall define an open space configured to receive the sample block.
7. The system of claim 1, wherein the one or more ejector mechanisms comprise a spring biased mechanism.
8. The system of claim 7, wherein the one or more ejector mechanisms comprise a housing at the spring biased mechanism.
9. An adaptor for supporting a sample holder relative to a sample block in a thermal cycler system, the adaptor comprising: a deck portion defining a plurality of apertures, the deck portion configured to support a sample holder; a peripheral sidewall extending downwardly from the deck portion; and one or more openings in the deck portion, wherein the deck portion and peripheral sidewall define an open space configured to receive the sample block of the thermal cycler system, and wherein, in a state of the sample block of the thermal cycler system received by the adaptor: the one or more openings of the adaptor are configured to permit passage of one or more ejector mechanisms of the thermal cycler system to extend through the one or more openings and into contact with the sample holder supported by the adaptor.
10. The adaptor of claim 9, wherein in the state of the sample block of the thermal cycler system received by the adaptor, the peripheral sidewall extends from the deck portion to proximate a base of the sample block.
11. The adaptor of claim 9, wherein the one or more openings extend through the peripheral sidewall to a surface of the deck portion.
12. The adaptor of claim 11, wherein: in the state of the sample block received by the adaptor, the peripheral sidewall of the adaptor protects the sample block from exposure to air.
13. The adaptor of claim 9, wherein the peripheral sidewall is configured to be received in a gap defined between a drip pan of the thermal cycler system and the sample block.
14. The adaptor of claim 9, wherein, in the state of the sample block of the thermal cycler system received by the adaptor, the plurality of apertures of the deck portion are arranged to be respectively aligned with a plurality of cavities in the sample block.
15. The adaptor of claim 9, wherein the one or more ejector mechanisms are positioned adjacent

one or more sides of the sample block, and wherein the one or more ejector mechanisms are configured to cause force to be exerted on the sample holder in a direction away from the sample block in a state of the sample holder received by the sample block.

16. The adaptor of claim 9, wherein the adaptor is configured to support the sample holder so as to place the sample holder in thermal contact with the sample block in the state of the sample block received by the adaptor.

17. The adaptor of claim 16, wherein the adaptor is configured to support the sample holder such that the plurality of apertures are positioned to respectively receive a plurality of wells of the sample holder.

18. The adaptor of claim 9, wherein the one or more ejector mechanisms comprise a spring biased mechanism, and wherein the one more ejector mechanisms comprise a housing at the spring biased mechanism.
